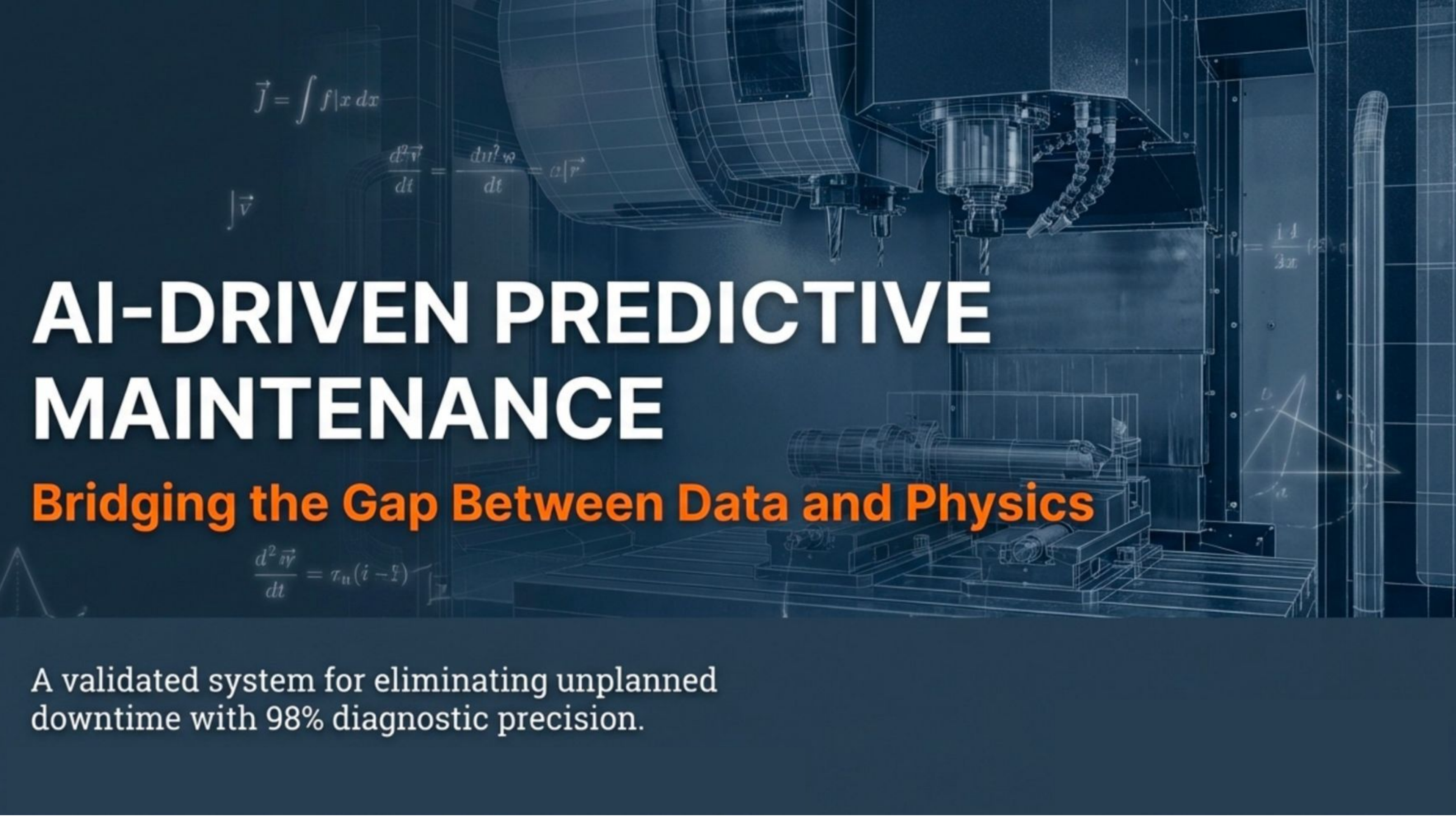




AI-DRIVEN PREDICTIVE MAINTENANCE

Bridging the Gap Between Data and Physics

A validated system for eliminating unplanned
downtime with 98% diagnostic precision.

Our Agenda

- 1. Introduction**
The Industrial Challenge
- 2. The Problem**
A Needle in a Haystack
- 3. The Solution**
An Iterative, Four-Sprint Journey
- 4. Key Results**
The Precision Breakthrough
- 5. Live Demo**
The Interactive Dashboard
- 6. Conclusion**
From Reactive to Proactive

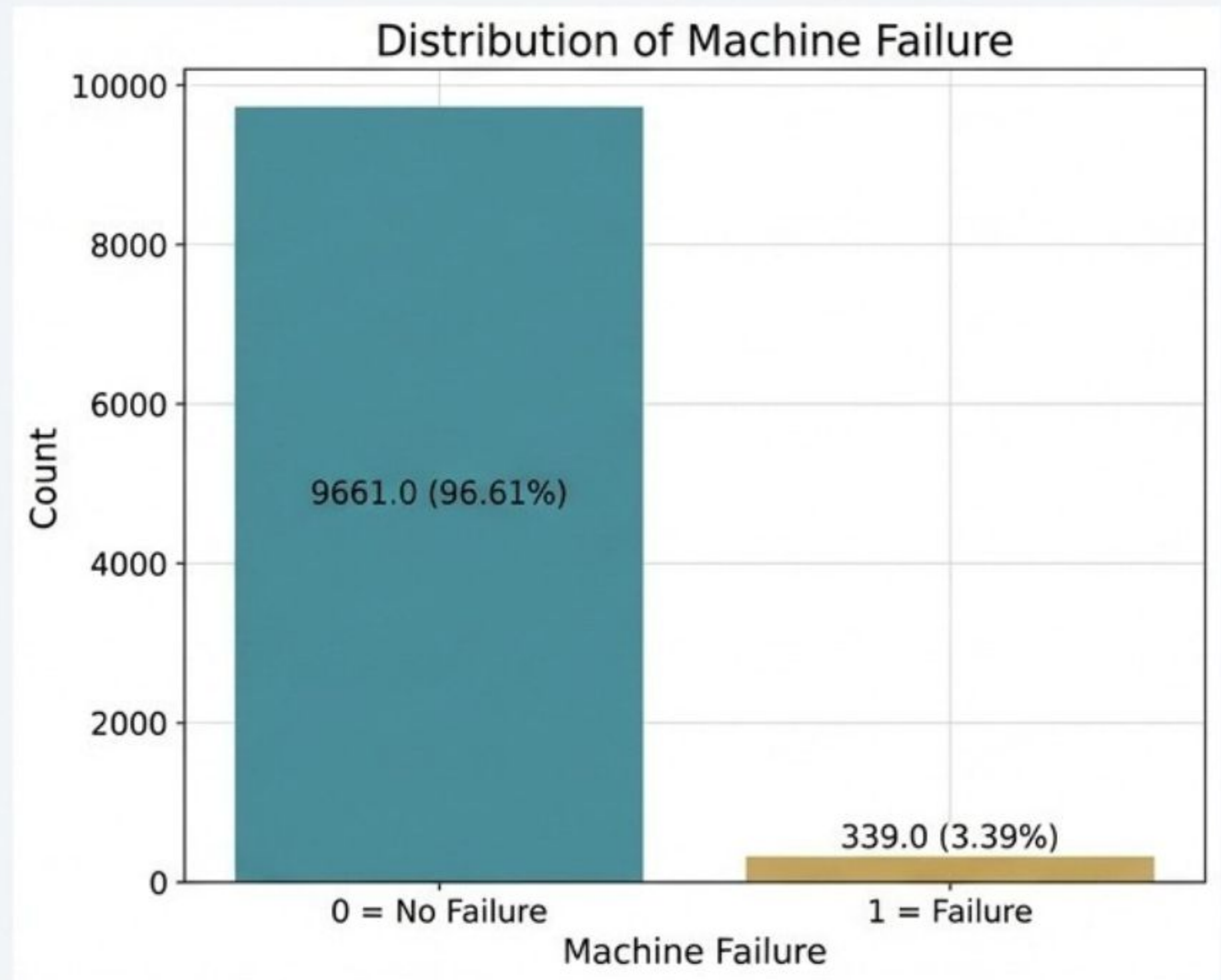
The High Cost of Unplanned Downtime

- Industry 4.0 demands maximum operational uptime, as a single hour of downtime can cost over twenty thousand dollars.
- Traditional maintenance is a trade-off:
 - **Reactive:** 'Run-to-failure' is costly and unsafe.
 - **Preventive:** Calendar-based replacement is inefficient, with **45%** of parts discarded while still functional.
- The Goal: A predictive system that enables maintenance *only when necessary*.



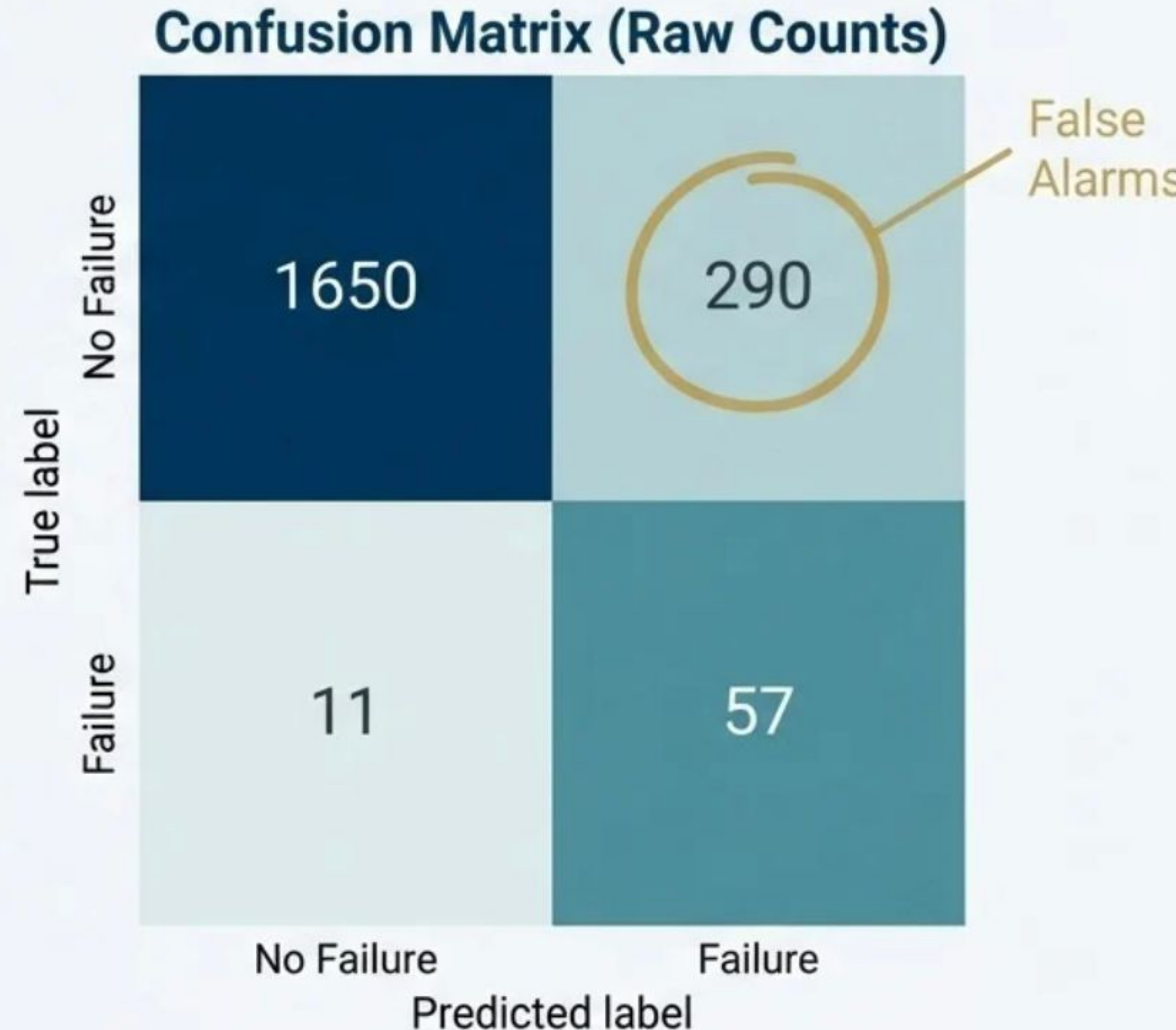
The “Needle in a Haystack” Problem

- The core technical challenge is extreme Class Imbalance.
- In our dataset, 96.61% of operations are normal.
- Only **3.39%** of data points represent a true machine failure.
- A naive model could achieve >96% accuracy by simply guessing 'No Failure' every time, making it useless in practice.

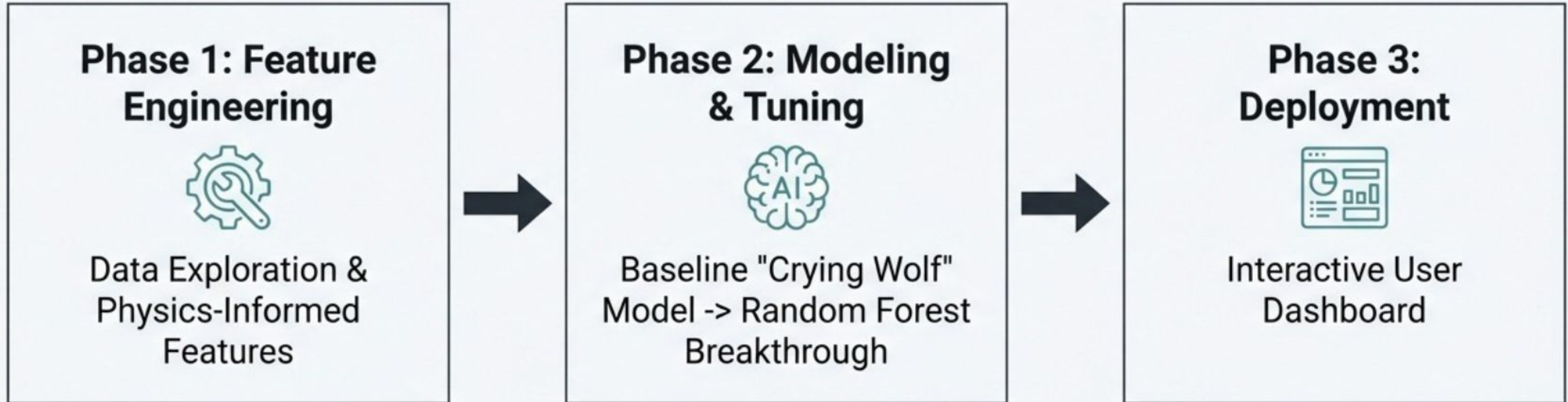


The Baseline Model: Crying Wolf

- Our initial model (Logistic Regression) achieved high **Recall (84%)**, meaning it found 57 out of 68 real failures.
- However, it suffered from extremely low **Precision (17%)**.
- This resulted in **290 false alarms** for only 57 correctly identified failures.
- This creates a 'Trust Deficit': a system that is constantly wrong will be ignored by technicians.



A Four-Sprint Journey to a Trusted Solution



- **Sprint 1: The Foundation**
- **Sprint 2: The Baseline**
- **Sprint 3: The Breakthrough**
- **Sprint 4: The Product**

The Secret Sauce: Teaching the Model Physics

- We hypothesized that raw sensor data was not enough. The model needed to understand engineering principles.
- We engineered two critical features from the raw inputs:
 - **Power (W)**: A direct measure of workload, combining Torque and Speed.
 - **Overstrain (min·Nm)**: A measure of accumulated stress, combining Tool Wear and Torque.

Power (Workload)

$$\text{Power [W]} = \frac{\text{Torque [Nm]} \times \text{Rotational Speed [rpm]}}{9.5488}$$

Overstrain (Accumulated Stress)

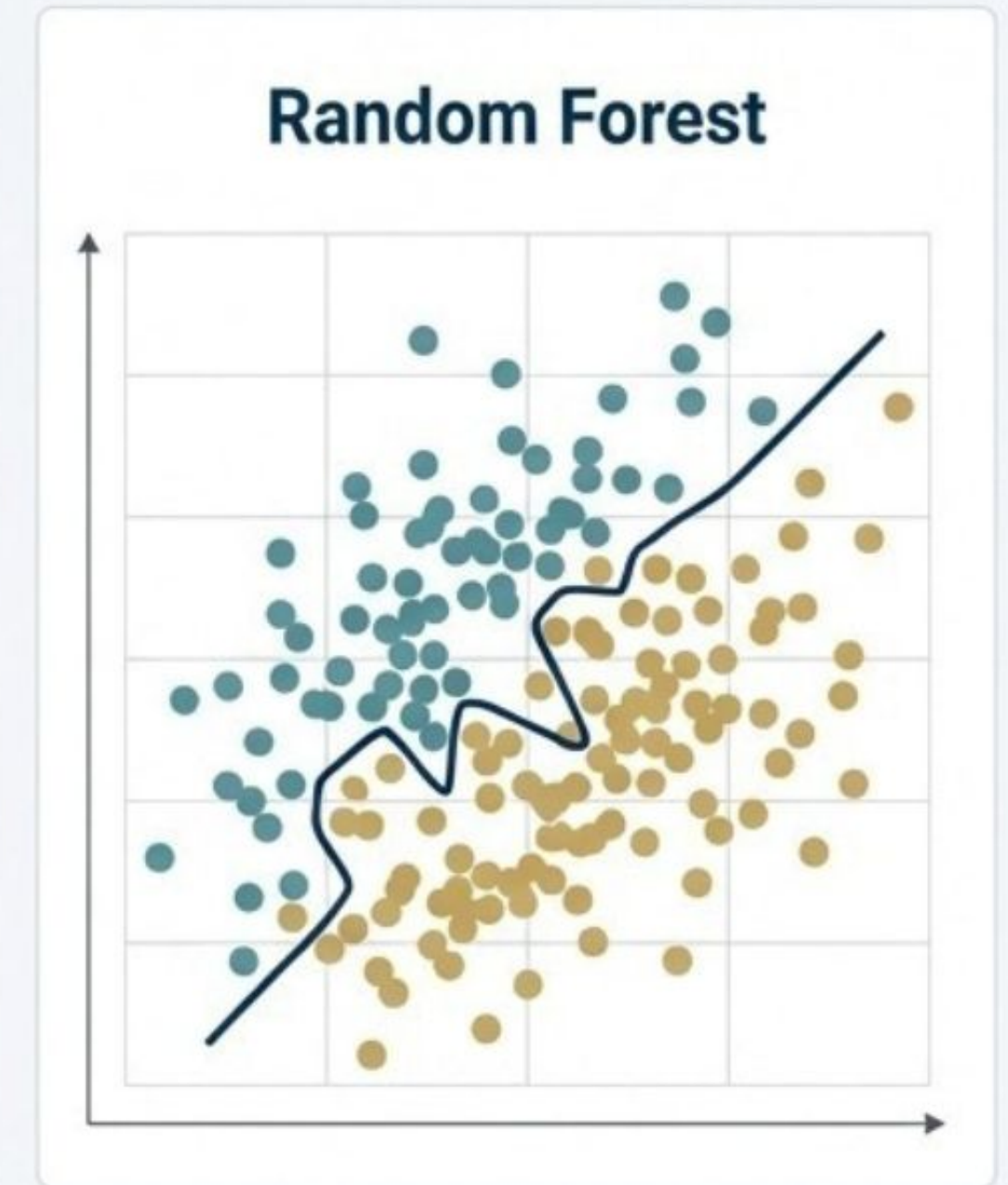
$$\text{Overstrain [min·Nm]} = \text{Tool Wear [min]} \times \text{Torque [Nm]}$$

From a Single Opinion to an Expert Committee

- The baseline model used Logistic Regression, which seeks simple, linear relationships.
- Failures are often non-linear (e.g., high rotational speed is only a risk when combined with extremely low torque).
- We switched to a **Random Forest** model.
- This is an 'ensemble' method that combines the predictions of hundreds of decision trees, effectively filtering out noise and capturing complex interactions.



Simple, Linear Boundary



Complex, Ensemble Boundary

The Breakthrough: Precision Solved

- The new strategy fundamentally solved the 'Trust Deficit'.
- False alarms on the test set dropped from **290 to just 1**.
- The system now delivers alerts that technicians can trust and act on immediately.

Metric	Baseline Model (Sprint 2)	Final Model (Sprint 3)
Precision	17%	98%
Recall	84%	82%
False Alarms	290	1

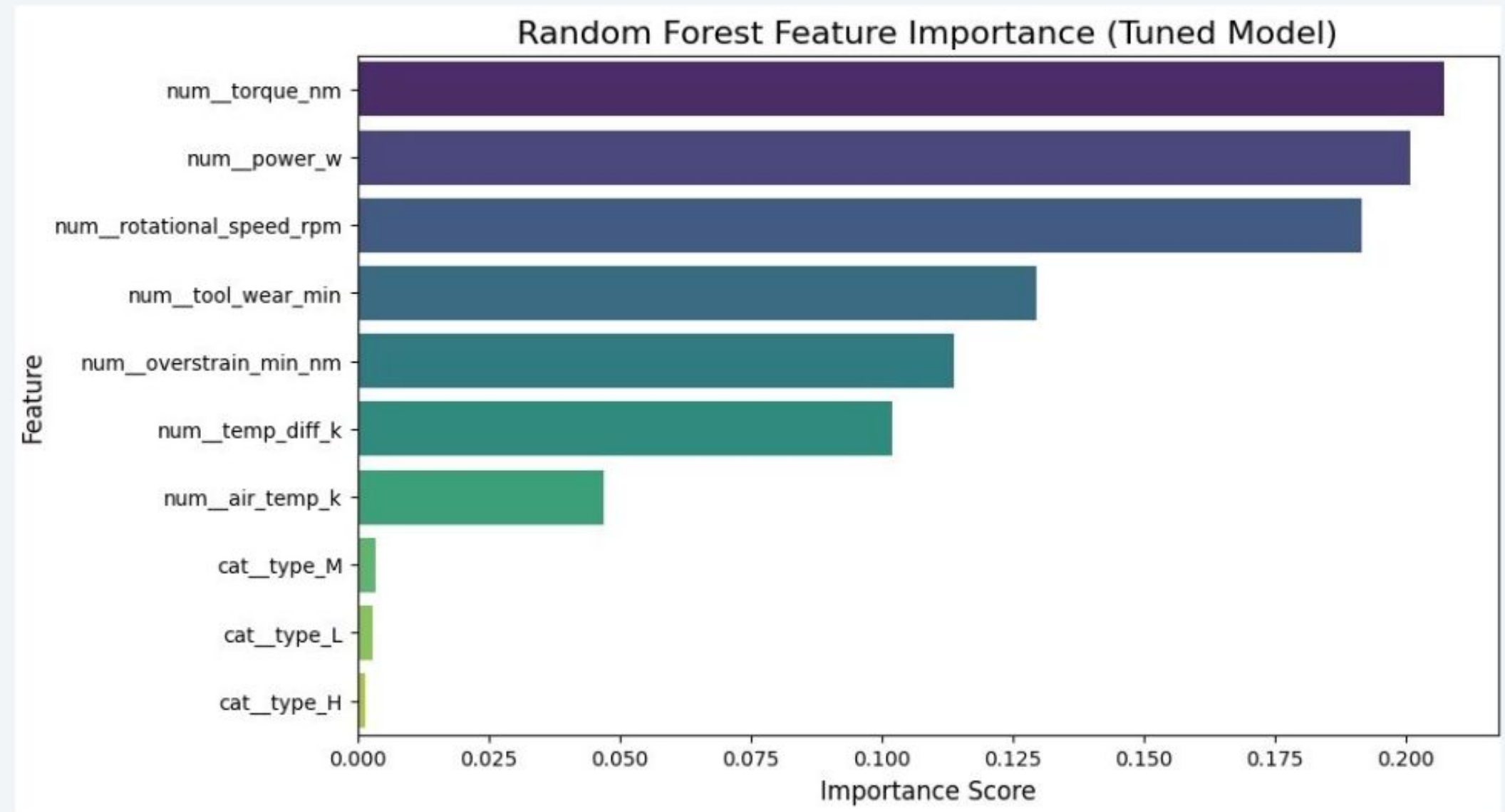
Validation: Why It worked

- * We analyzed the final model to understand its decision-making logic.

- * The model confirmed our hypothesis: physics-based features were paramount.

- * Torque, our engineered Power feature, and Rotational Speed were the top three most important predictors of failure.

- * This proves that building the model on physical laws works better than relying only on raw data.



Beyond Prediction: Pinpointing the Root Cause

- Predicting failure is only half the battle. Technicians need to know *why*.
- We developed a second model, a **Multi-Output Classifier**, for diagnostics.
- This model identifies the specific failure type (e.g., Power Failure, Overstrain).
- Achieved **98.95% diagnostic accuracy**, turning a simple warning light into a precise diagnostic tool.



Power Failure



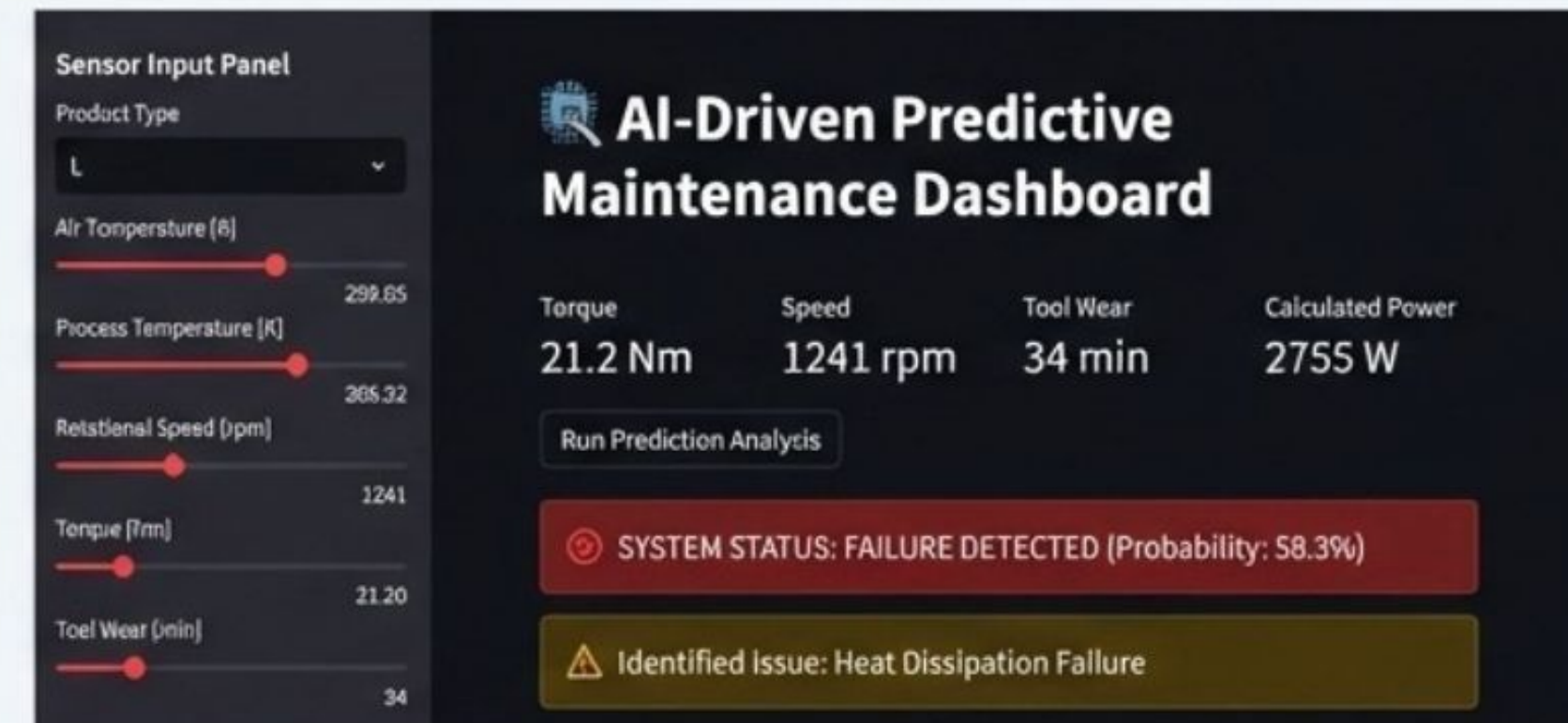
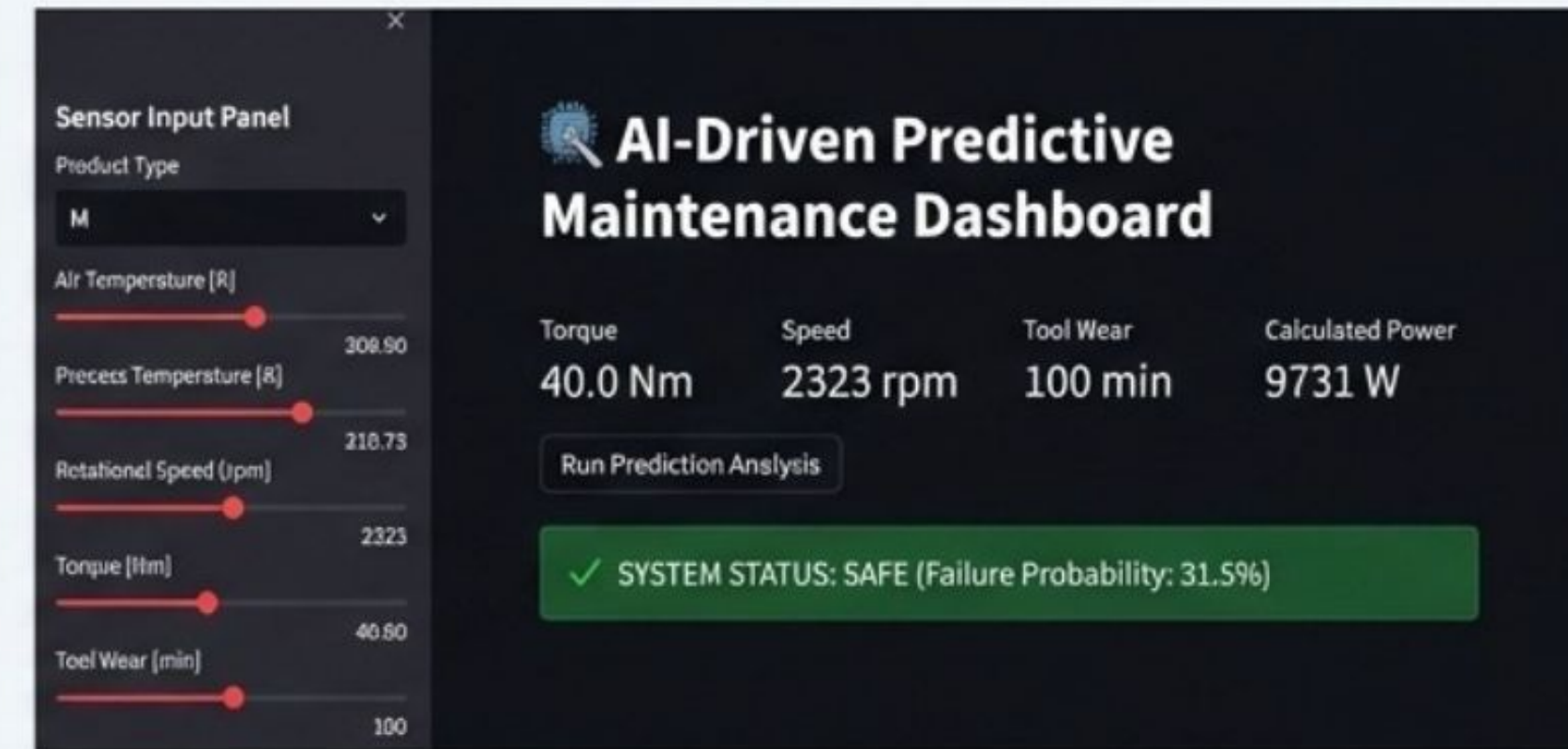
Overstrain Failure



Heat Dissipation Failure

The Final Product: An Interactive Dashboard

- We deployed our models into a user-friendly Streamlit application.
- **Real-Time Status:** Clear green ('Safe') and red ('Failure Detected') banners visible from a distance.
- **Scenario Simulation:** Operators can use sliders to ask 'what-if' questions and understand operational limits.
- **Instant Diagnosis:** When a failure is predicted, the root cause is displayed immediately.



Recommendations for Implementation



Phase 1: Pilot Integration

Deploy the dashboard on a single, high-value production line for a 90-day trial.



Phase 2: Human-in-the-Loop

Implement a feedback mechanism allowing technicians to confirm or correct the model's diagnosis, providing data for continuous improvement.



Phase 3: ERP & Automation

Integrate alerts with the plant's ERP system to automate work orders and spare parts management.

THANK YOU